Nikky Amresh

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PROFESSIONAL SUMMARY

Senior SDE, Platforms Engineering at Educational Initiatives (Mindspark), owning backend services for an adaptive learning platform used by 500,000+ students globally. 4+ years full-time across edtech, SaaS, and developer tooling. Depth in Node.js/TypeScript and PHP microservices, API design, performance engineering, and AWS, owning features end to end from design doc through production health.

CORE COMPETENCIES

Backend & API Design | Microservices | Node.js & TypeScript | Performance Optimization | System Design | AWS & Docker

WORK EXPERIENCE

Educational Initiatives (Mindspark)

Dec 2023 - Present

Senior SDE, Platforms Engineering · Bangalore, India · Hybrid

- Promoted twice in three years (SDE 2 in July 2025, **Senior SDE in July 2026**) on the platform backend team behind Mindspark, an adaptive learning product used by **500,000+ students** globally.
- Owned the backend for the Computational Thinking product (Amazon Feature Engineering partnership), shipping to **100,000+ students**.
- Cut QT report generation from **2m 16s to 18.9s (86% reduction)** through backend refactoring and query optimization.
- Built Mira Platform microservices (auth APIs, SMS notifications, approval workflows, event scheduling) on a shared error-handling framework, and designed the Pre-Post Assessment System with impact tracking and permissions.
- Built internal **MCP (Model Context Protocol) servers** for Jira triage, PR review, commit analysis, and Confluence operations, plus the MCP Mindspark Control Panel for order and org lifecycle management.

Stack: PHP, Node.js, TypeScript, Python, MongoDB, AWS (EC2, S3), Docker, Microservices

Wing Assistant

Jun 2023 - Dec 2023

PHP Developer · California, United States · Remote

- Built real-time group and personal chat for two web portals with schema and query design tuned for low-latency delivery; added Pipedrive CRM and secure webhook integrations; resolved Sentry-tracked production issues. PHP, JavaScript, MySQL, MongoDB, Docker, Sentry

HackerEarth

Apr 2022 - May 2023

Full Stack Engineer · Bangalore, India · Hybrid

- Delivered **40+ full-stack projects** (Django, Laravel, MERN, MEVN, Express, TypeScript, Ruby on Rails) and 20+ DevOps assessment scenarios (Docker, Linux, Bash) for the technical assessment platform; led knowledge-transfer sessions.

Gramoday

Sep 2021 - Dec 2021

Backend Developer, Internship · Noida, India · Remote

- REST APIs with Express.js, MongoDB, and TypeScript for a mobile app; Bitbucket CI/CD pipelines and AWS S3 deploy scripts.

Fotonicia

Sep 2019 - Jul 2021

Software Engineer, Internship · New Delhi, India · Remote, full-time scope

- Magento PHP plugins, paywall modules, and marketplace features for e-commerce; led MEVN implementation; REST APIs and AWS CI/CD pipelines.

PROJECTS

Astra - local-first AI task manager: parses plain English into structured tasks, auto-links tasks, people, and goals into a knowledge graph, FTS5 search, two-way Google Calendar sync. Node.js, TypeScript, SQLite + FTS5, Groq/Claude, Vue.js · github.com/NikkyAmresh/astra

Icex - LeetCode practice extension, **live on the VS Code Marketplace**: 40+ commands, Firestore cloud sync, custom file formats and editors, Chrome companion. TypeScript, VS Code API, Firestore · github.com/NikkyAmresh/Icex

Storyboard - five-stage AI comic and video pipeline holding state across ComfyUI, FFmpeg, LLM APIs, and Instagram's Graph API, with runtime workflow patching. Node.js, Express + SSE, FFmpeg, SQLite · github.com/NikkyAmresh/nodeimg

Also mext (VS Code PostgreSQL client, SQL panels, JWT inspectors), WorldBuilder (UE5 C++ procedural roads and buildings), Algolify (interactive DSA lessons).

EDUCATION & ACHIEVEMENTS

B.Tech, Computer Science Engineering · Roorkee Institute of Technology — 2019 - 2023 · 8.8 CGPA

Smart India Hackathon 2022 Grand Finalist (Verifi, offline OTP auth for DRDO) · NECTF organizer, international CTF, 400+ participants from 25+ countries · regional-level chess player.

SKILLS

Languages: JavaScript, TypeScript, Python, PHP, C++
Backend & Data: Node.js, Express.js, FastAPI, Flask, PHP · MongoDB, MySQL, PostgreSQL, SQLite + FTS5, Firebase
Architecture & Cloud: Microservices, event-driven, API design, real-time (WebSockets, WebRTC, HLS) · AWS (EC2, S3, Lambda), Docker, Linux, CI/CD (GitHub Actions, Bitbucket), Sentry
AI/LLM & Frontend: Groq, Claude, OpenAI, LangChain, vector databases, MCP · React.js, Vue.js